

Static analysis with focus on using language models and Hoare logic

Rashid Harvey
Freie Universität Berlin
IoT & Security Seminar Report

Abstract—Abstract

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

A. IoT Security

- little resources for Security - custom-built software - limited resources - difficult to update

B. Static Analysis

Static analysis is the analysis of code, binaries or other artifacts to check certain properties without, in contrast to dynamic analysis, executing any code.

Examples for static analysis: linting, formatter, syntax checking, type checking, flow analysis

Examples for dynamic analysis: tests, emulation

Advantages of static analysis - high efficiency, run quickly, cheap - Find errors quickly

Disadvantages of static analysis - theoretical limitations (Halteproblem) - high possible false-positive rate

C. Large Language Models for Coding

LLM have seen massive advancements for coding and security. For example, looking at the "SWE-bench Verified" benchmark of 500 human-selected GitHub Issues the best-performing model, Claude 2, was only able to solve 1,96% of the issues at the time of the benchmarks inception in 2023. [?]. This stands in stark contrast to the current state-of-the-art: Claude Fable 5 solves 95% of the same benchmark at the time of the models release. [?]

Even if one was to account for the limited effectiveness of public benchmarks, this shows quite clearly that LLMs can effectively be used for developing code. ¹

The US government also seems to agree. On the 12th of June 2026 they forced Anthropic to take down Fable 5, the already down-graded safe-guarded version of Mythos 5, from public access after just about 76 hours by imposing a national-security export control. [?]

Earlier, AISLE had already found several major security issues in famously hard-audited, mature codebases like OpenSSL. [?]

¹For example, the benchmarks might be leaky, [?], and the models train and memorize the solutions instead of learning the general principles. [?]

II. STATIC ANALYSIS METHODS

A. Lexical Methods and Methods Using the Abstract Syntax Tree Representation

Lexical methods and methods using the abstract syntax tree (AST) representation are pretty "simple" methods, you would often summarize as linting.

Lexical analysis simple means breaking down text into the smallest semantic units "tokens" and analyzing them. The AST is a representation of the code as a tree, which allows for more depth. The AST is what a compiler, transpiler, interpreter and most linters and formatters will consume.

Examples

- 1) Syntax errors
- 2) Name errors or misspellings
- 3) Missing definitions and declarations
- 4) Rule matching, e.g. leaked secrets

B. Semantic Methods

1) *Type checking*: The idea behind types is that any data on a digital computer is simply some bitstring. The interpretation of this bitstring is what makes the data a value.

The data only makes sense if we define operations on it, but these operations depend on the type. For example, in Python the + operator for integers does integer addition, but for strings it does string concatenation. However, if you try to "add" an integer with a string, you get a type error. In Python type checking is done at runtime and is therefore called "dynamic type checking". However, for static analysis, we can check the types ahead of run-time "static type checking". [?], [?]

C. Flow Control

- info flow control - control flow analysis - taint analysis

D. Case study: Rust

Rust is a low-level programming language started in 2006 by Graydon Hoare as a side project, then officially sponsored by Mozilla in 2009 and it finally reached maturity (v1.0) in 2015. Rust was specifically created to solve the flaws of C/C++. [?], [?] This section tries to summarize how this is done in Rust.

"Who doesn't want memory safety, *and* fast performance, *and* a friendly compiler, *and* great tooling, among a host of other wonderful features?" [?]

The Rust language has a very strict compiler which includes not just type checking but also a borrow checker and more.

[?] These checks lead to the conclusion that regular Rust code compiled through this strict compiler is also "safe": "If it compiles, it works".

"Rust's type system and runtime guarantee the absence of data races, buffer overflows, stack overflows, and accesses to uninitialized or deallocated memory." [?]

1) *Case studies:* Large-scale industry data supports this: Microsoft reports that roughly 70% of its annually assigned CVEs are memory-safety issues, [?], [?], while Android saw memory-safety vulnerabilities fall from 76% to 24% after shifting new development to memory-safe languages. [?]

III. FORMAL METHODS

A. Introduction to Formal Methods

classical static analysis finds symptoms; formal methods prove properties

When applying formal methods first a formal model is created. The kind of model used depends on the use case After the model is set up, model checking must be applied, this is called formal verification.

The beauty of Hoare logic

B. Overview of Methods

- 1) Model-based formal specification languages like Z, Vienna Development Model (VDM), and B
- 2) Interactive proof assistants like Rocq (formerly Coq), [?] Isabelle/HOL [?] or Lean
- 3) Verification-aware programming languages like Dafny, [?] SPARK, F*, [?] Why3, Viper, Whiley or hax

C. Examples

1) *Dafny:* In the original paper on Dafny, Leino gave the challenges of directly using interactive proof assistants and satisfiability-modulo-theories (SMT) solvers as main reason for the creation of Daphny. The idea is that the programmer doesn't have to interact with any proof assistant or solver directly, only make certain annotations [?]. Dafny compiles down to C, Java, JavaScript, Go, and Python

D. Limitations

1. Gaps: Scalability, Real-Time Performance, Cooperation between Formal Methods, Adaptability, Lack of Focus on Security & Privacy ?, Usability,

IV. DISCUSSION

-¿ build "theoretical emulators"

V. CONCLUSION

REFERENCES

- [1] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan, "SWE-bench: Can language models resolve real-world github issues?" in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024, baseline: best model (Claude 2) solves 1.96% of the issues.
- [2] Anthropic, "Claude fable 5 and claude mythos 5 system card," <https://www-cdn.anthropic.com/d00db56fa754a1b115b6dd7cb2e3c342ee809620.pdf>, 2026, released 9 June 2026. SWE-bench Verified: Fable 5 95.0%, Mythos 5 95.5%, Opus 4.8 88.6%.
- [3] R. Aleithan, H. Xue, M. M. Mohajer, E. Nnorom, G. Uddin, and S. Wang, "SWE-Bench+: Enhanced coding benchmark for LLMs," *arXiv preprint arXiv:2410.06992*, 2024, audit of SWE-bench: 32.67% of resolved instances had solution leakage, 31.08% passed via weak tests; filtering dropped SWE-Agent+GPT-4 from 12.47% to 3.97%. Same issues in Lite/Verified.
- [4] S. Liang, S. Garg, and R. Zilouchian Moghaddam, "The SWE-bench illusion: When state-of-the-art LLMs remember instead of reason," *arXiv preprint arXiv:2506.12286*, 2025, memorization/contamination: models identify buggy file paths up to 76% from issue text alone vs. 53% for out-of-benchmark repos.
- [5] Anthropic, "Statement on the US government directive to suspend access to fable 5 and mythos 5," <https://www.anthropic.com/news/fable-mythos-access>, Jun. 2026, export-control directive received 12 June 2026, 5:21pm ET; Fable 5/Mythos 5 launched 9 June 2026 (roughly 72 h earlier).
- [6] AISLE, "AISLE discovered 12 out of 12 OpenSSL vulnerabilities," <https://aisle.com/blog/aisle-discovered-12-out-of-12-openssl-vulnerabilities>, Jan. 2026, aI-driven discovery of 12 CVEs (incl. CVE-2025-15467, HIGH, CVSS 9.8) in a mature, heavily audited codebase; some bugs date back to 1998. AISLE's system was powered by Claude Opus 4.6.
- [7] B. C. Pierce, *Types and Programming Languages*. Cambridge, MA: MIT Press, 2002, standard reference. Defines a type system as a static method; contrasts static type checking with dynamic typing.
- [8] Python Software Foundation, "The python language reference: Data model," <https://docs.python.org/3/reference/datamodel.html>, 2026, official source for runtime/dynamic typing: every object has a type; operators desugar to type-dependent special methods, e.g. `a+b` calls `type(a).__add__ / type(b).__radd__`.
- [9] C. Thompson, "How Rust went from a side project to the world's most-loved programming language," <https://www.technologyreview.com/2023/02/14/1067869/rust-worlds-fastest-growing-programming-language/>, Feb. 2023, single source covering the full origin sentence: 2006 side project by Hoare, Mozilla official sponsorship 2009, v1.0 in 2015, motivated by C/C++ memory-safety flaws.
- [10] G. Hoare, "10 years of stable rust: An infrastructure story," <https://rustfoundation.org/media/10-years-of-stable-rust-an-infrastructure-story/>, May 2025, guest post by Rust's initial author. Primary source for the origin story: project begun 2006, "near-decade of development" before the 1.0 release (15 May 2015). Mozilla sponsorship from 2009.
- [11] S. Klabnik and C. Nichols, *The Rust Programming Language*, 3rd ed. No Starch Press, 2025, official book; chapters on ownership and borrowing. Freely available: <https://doc.rust-lang.org/book/>.
- [12] N. D. Matsakis and F. S. Klock II, "The Rust language," in *Proceedings of the 2014 ACM SIGAda Annual Conference on High Integrity Language Technology (HILT '14)*, ser. Ada Letters, vol. 34, no. 3. ACM, 2014, pp. 103–104, source of the quote on Rust's guarantees (absence of data races, buffer/stack overflows, use of uninitialized/deallocated memory).
- [13] M. Miller, "Trends, challenges, and strategic shifts in the software vulnerability mitigation landscape," Presentation at BlueHat IL 2019; slides at <https://github.com/microsoft/MSRC-Security-Research>, Microsoft Security Response Center, Feb. 2019, primary source for the roughly 70% figure: roughly 70% of the CVEs Microsoft assigns each year are memory-safety issues, stable for 12+ years.
- [14] Microsoft Security Response Center, "We need a safer systems programming language," <https://msrc.microsoft.com/blog/2019/07/we-need-a-safer-systems-programming-language/>, Jul. 2019, mSRC explicitly argues for memory-safe languages such as Rust; roughly 70% of CVEs are memory-safety issues despite code review, training, and static analysis.
- [15] J. Vander Stoep and A. Rebert, "Eliminating memory safety vulnerabilities at the source," <https://security.googleblog.com/2024/09/eliminating-memory-safety-vulnerabilities-Android.html>, Sep. 2024, case study: Android memory-safety vulnerabilities fell from 76% (2019) to 24% (2024) over 6 years as new development shifted to memory-safe languages, well below the roughly 70% industry norm.

- [16] The Rocq Development Team, “The rocq prover (formerly coq), version 9.0,” <https://rocq-prover.org/>, 2025, renaming Coq to The Rocq Prover; first release under the new name (v9.0, 12 March 2025).
- [17] K. R. M. Leino, “Dafny: An automatic program verifier for functional correctness,” in *Logic for Programming, Artificial Intelligence, and Reasoning (LPAR-16)*, ser. Lecture Notes in Computer Science, E. M. Clarke and A. Voronkov, Eds., vol. 6355. Springer, 2010, pp. 348–370.
- [18] N. Swamy, C. Hrițcu, C. Keller, A. Rastogi, A. Delignat-Lavaud, S. Forest, K. Bhargavan, C. Fournet, P.-Y. Strub, M. Kohlweiss, J.-K. Zinzindohoué, and S. Zanella-Béguelin, “Dependent types and monadic effects in F^* ,” *ACM SIGPLAN Notices (POPL 2016)*, vol. 51, no. 1, pp. 256–270, 2016.